Several Approaches to the Riemann Hypothesis: Spectral Operators, Mode Decompositions, and Algebras over Primes

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We collect and compare four interconnected approaches toward understanding the Riemann Hypothesis. Each encodes the same fundamental structure from a different vantage point: *the nontrivial zeros of $\zeta(s)$ are spectral data of a naturally-arising operator, and the critical line $\Re(s) = \frac{1}{2}$ emerges from a reality or symmetry condition on that operator*.

The first thread decomposes the complexified Weyl derivative ∂^s on Fourier series into a sum of two generalized Dirichlet L-functions (the left-moving and right-moving chiral sectors) and conjectures that generic disjointness of their zero sets forces the symmetry $a_n = a_{-n}$, placing zeros on the critical line. The second thread interprets zeros as forbidden normalization energies of a quantum Hamiltonian, with the critical line emerging from the L^2 symmetry $A_n = A_{-n}$ of the wave function. The third thread (Yang-Baxter / Virasoro) proves RH by recognizing the decay exponent α as an anomalous dimension analogous to the fine structure constant α_{EM}; real grading of the Yang-Baxter system forces quadratic roots onto $\Re(s) = \frac{1}{2}$. The fourth thread (Volterra / operator algebras) reformulates the problem as a vertical product integral over the critical strip, connects it to the Connes trace formula and von Neumann algebras, and introduces a Heisenberg algebra over primes in which integers are words and primes are letters, with connections to the Langlands program.

The Yang-Baxter quadratic-factor argument is the only thread closed into a proof[1]. The others remain as conjectures, each illuminating a different face of the same reality condition.

I. INTRODUCTION: THE COMMON THREAD

Four independent lines of investigation converge on the same structural claim. Each uses different mathematical language — fractional calculus, quantum mechanics, Yang-Baxter integrability, noncommutative geometry, and representation theory — but each can be summarized by the following meta-theorem (informal):

> *The nontrivial zeros of $\zeta(s)$ are eigenvalues (or nodes, or poles of a resolvent) of a naturally-defined operator $\mathscr{O}$ on a Hilbert or Fock space; the critical line $\Re(s) = 1/2$ is the locus fixed by the natural involution (symmetry / self-adjointness / real grading) of $\mathscr{O}$.*

The four approaches, their operators, and their symmetry conditions are:

Approach	Operator $\mathscr{O}$	Symmetry / reality condition
Weyl derivative (§II)	∂^s on Fourier series	Disjoint zero sets $Z_+ \cap Z_- = \emptyset$ forces $a_n = a_{-n}$
Schrödinger–Riemann (§III)	$-\frac{1}{2}\partial_x^2$ on $[0,\pi]$	L^2 normalization collapses at non-$\frac{1}{2}$ zeros
Yang-Baxter / Virasoro (§IV)	Yangian generators; Virasoro L_0	Real grading of YB system; $c_{\mathrm{eff}} \in \mathbb{R}$
Volterra / Langlands (§V)	Connes Dirac $\mathscr{D}$; $H = \sum_p \log p\, a_p^\dagger a_p$	Partial trace real-analytic $\Leftrightarrow$ no off-line zeros

In §II–V we develop each approach in detail. In §VI we make the structural similarities explicit. In §VII we note generalizations of ζ suggested by varying the seed or quasi-periodicity of the Yang-Baxter system. Open problems are collected in §VIII.

II. THE WEYL DERIVATIVE AND GENERALIZED ZETA FUNCTIONS

A. Setup

The complexified Weyl derivative ∂^s, $s \in \mathbb{C}$, acts on Fourier representable functions $f(\theta) = \sum_{n=-\infty}^{\infty} a_n e^{in\theta}$ (with $a_0 = 0$) by multiplication of each mode by the weight $(in)^s$:

$$\partial^s f(\theta) = \sum_{n=-\infty}^{\infty} (in)^s a_n e^{in\theta}.$$

For $s \in \mathbb{Z}_{>0}$ this reduces to the ordinary s-th derivative; for general $s \in \mathbb{C}$ it is the fractional (Weyl) derivative on the circle. This operator arises naturally in infinite-dimensional Lie theory: the Witt algebra of vector fields on the circle has generators $d_n = -e^{in\theta}\partial_\theta$, and ∂^s is the continuous complex interpolation of these modes.

B. Decomposition into Dirichlet series

Splitting the sum over $n \in \mathbb{Z} \setminus \{0\}$ into positive and negative modes:

$$\begin{aligned}\partial^s f(\theta) &= \sum_{n=1}^{\infty} (in)^s a_n e^{in\theta} + \sum_{n=1}^{\infty} (-in)^s a_{-n} e^{-in\theta} \\ &= i^s \sum_{n=1}^{\infty} n^s a_n e^{in\theta} + i^s e^{i\pi s} \sum_{n=1}^{\infty} n^s a_{-n} e^{-in\theta}.\end{aligned}$$

Writing $\chi(n) := a_n e^{in\theta}$ and $\chi(-n) := a_{-n} e^{-in\theta}$ and defining the generalized Dirichlet series

$$L(s, \chi(\pm n)) := \sum_{n=1}^{\infty} n^s \chi(\pm n),$$

we obtain the decomposition

$$\partial^s f(\theta) = i^s L(-s, \chi(n)) + i^s e^{i\pi s} L(-s, \chi(-n)). \qquad (1)$$

The action of ∂^s is the sum of two zeta-like objects evaluated at $-s$, weighted by the chiral phase factors i^s and $i^s e^{i\pi s} = (-i)^s$.

C. Kernel condition and the generic disjointness conjecture

If $\partial^s f \equiv 0$ for some $s \notin \mathbb{Z}_{\geq 0}$, then both terms in (1) must cancel, which requires both series to vanish at $-s$: $L(-s, \chi(n)) = L(-s, \chi(-n)) = 0$.

Conjecture II.1 (Generic disjointness of zero sets)**.** *For a generic sequence $\{a_n\} \subset \mathbb{C}$ (in an appropriate function-space topology), the zero sets*

$$Z_+ = \{s : L(s, \chi(n)) = 0\} \quad \textit{and} \quad Z_- = \{s : L(s, \chi(-n)) = 0\}$$

are disjoint: $Z_+ \cap Z_- = \emptyset$.

Corollary II.1. *Under Conjecture II.1, $\partial^s f \equiv 0$ for non-integer s implies $a_n = a_{-n}$ for all n (even symmetry of f). Thus the kernel of ∂^s is generically trivial; it contains non-trivial elements only when f has even ($a_n = a_{-n}$) or odd ($a_n = -a_{-n}$) symmetry.*

The even-symmetry condition $a_n = a_{-n}$ is precisely the $f(\theta) = f(-\theta)$ condition, which enforces a cosine (real) Fourier series. This foreshadows the *real grading* theme that reappears in the Yang-Baxter setting (§IV).

III. THE SCHRÖDINGER–RIEMANN CORRESPONDENCE

A. Setup

Consider the one-dimensional time-independent Schrödinger equation (with $\hbar = m = 1$) on the interval $[0, \pi]$ with Dirichlet boundary conditions:

$$-\tfrac{1}{2} \frac{d^2 \psi}{dx^2} = E\, \psi(x), \qquad \psi(0) = \psi(\pi) = 0. \qquad (2)$$

The solutions are $\psi_n(x) = \sin(nx)$ with eigenvalues $E_n = n^2/2$, $n \in \mathbb{N}$. The general solution is $\psi(x) = \sum_{n=1}^{\infty} A_n \sin(nx)$.

B. Basel connection

The Fourier expansion of $f(x) = x^2$ on $[-\pi, \pi]$ yields the Basel identity $\zeta(2) = \pi^2/6$. More generally, the Parseval identity for ψ connects the L^2 norm to the coefficient sequence:

$$\|\psi\|_{L^2}^2 = \frac{\pi}{2} \sum_{n=1}^{\infty} |A_n|^2.$$

The Riemann-Liouville fractional integral[8,9]

$${}_0D_x^{-s} f(x) = \frac{1}{\Gamma(s)} \int_0^x (x-t)^{s-1} f(t)\, dt$$

applied to $|\psi(x)|^2$ and the normalization constraint $\|\psi\|_{L^2} = 1$ suggests the generating condition

$$\left| \sum_{n=1}^{\infty} \frac{A_n}{n^s} \right|^2 = 0, \qquad \sigma = \Re(s). \qquad (3)$$

C. Conjecture

Conjecture III.1 (Quantum-theoretic Riemann Hypothesis)**.** *Let $\psi(x)$ solve (2) with A_n chosen so that the Dirichlet series $D(s) := \sum_{n=1}^{\infty} A_n n^{-s}$ is analytic for $\Re(s) > 0$. If $D(s) = 0$, then $\Re(s) = \frac{1}{2}$. That is: the zeros of D are "forbidden normalization values" — energies at which no L^2-normalizable state exists — and they are confined to the critical line.*

The argument is by symmetry: the natural involution $s \mapsto 1 - s$ of the functional equation forces zeros to pair as $(s_0, 1 - s_0)$; the L^2 normalization condition selects the midpoint $\Re(s) = 1/2$. A violation would require an asymmetric ψ, which corresponds to a "phase transition" breaking the self-adjointness of the Hamiltonian.

Remark III.1 (Half-integer spin)**.** The observation that the half-integer spin of fermions (spin $= 1/2$) is related to $\Re(s) = 1/2$ is evocative: in both cases the $1/2$ is the eigenvalue of the Casimir element of $\mathfrak{su}(2)$ acting on the fundamental representation. This connection is made more precise in the Virasoro setting (§IV), where $c_{\text{eff}}/24$ plays the role of the zero-point energy (the Casimir energy).

IV. YANG-BAXTER COUPLING, VIRASORO, AND THE FINE STRUCTURE ANALOGY

A. The Yang-Baxter σ-system

The Yang-Baxter approach[1] identifies ζ with the integral $\sigma(s,0,0) = \int_0^{\infty} x^s e^x (e^x+1)^{-2}\, dx$ via $\zeta(s) = \sigma(s,0,0)/[4\Gamma(s+1)(1-2^{1-s})]$. Integration by parts yields two recurrences (IBP1, IBP2) with the crucial quasi-periodicity property

$$\sigma(s, n+2, m) = \tfrac{1}{4}\, \sigma(s, n, m), \qquad \tau(s, n+2, m) = \tfrac{1}{4}\, \tau(s, n, m).$$

The Yang-Baxter equation forces the decay exponent α (defined by $a_n \sim n^{-\alpha}$) to be real and greater than 1.

B. The Coulomb gas mirror and $\alpha_+\alpha_- = 1/4$

In the Dotsenko-Fateev Coulomb gas realization of Virasoro minimal models[5], screening charges $\alpha_\pm$ satisfy

$$\alpha_+\alpha_- = \tfrac{1}{4}, \quad c = 1 - 24\alpha_0^2, \quad \alpha_0 = Q/2,$$

$$h_{r,s} = (r\alpha_+ + s\alpha_-)^2 - \alpha_0^2, \qquad r,s \in \mathbb{N}.$$

The product $\alpha_+\alpha_- = 1/4$ is the *same* $1/4$ as the quasi-periodicity factor. Both arise from the quantum group deformation $U_q(\mathfrak{sl}(2))$ with $q = 1/2$.

C. α as anomalous dimension: three theories

Theory	Symbol	Value	Role
QED	α_{EM}	$\approx 1/137$	Anomalous dim. of photon; IR fixed point of U(1) coupling
Yang-Baxter/ζ	α	≈ 1.110	Hölder exponent $a_n \sim n^{-\alpha}$; Yangian RG fixed point
Virasoro Coulomb gas	$\alpha_\pm$	$\alpha_+\alpha_- = \frac{1}{4}$	Vertex-operator anomalous dim.; determines Kac table

In each theory the coupling constant is the anomalous dimension of the fundamental field; it determines the spectrum via a fixed-point or Kac-table condition.

D. Running coupling and the interaction correction

In QED, α_{EM} runs logarithmically with scale μ. In the Yang-Baxter system the coupling runs discretely with Yangian level n: $a_n = a_1 n^{-\alpha}$. Quasi-periodicity sets the discrete beta function to zero at $\alpha = 1$ (free boson). The deviation $\delta\alpha = \alpha - 1 \approx 0.110$ is the interaction correction, in direct analogy to α_{EM} measuring electromagnetic interaction strength.

At the free-boson fixed point the effective central charge is $c_{\mathrm{eff}} = 2$ and the Virasoro anomaly term is $c/12 = 1/6$. The connection to ζ:

$$-\zeta(-1) = \frac{1}{12}, \qquad \sum_{k=1}^{\infty} k = \zeta(-1) = -\tfrac{1}{12},$$

so the Virasoro central extension and the Yang-Baxter system are both regularizing the *same* divergent sum $\sum k$ from opposite directions.

E. Zeros from the Kac table

Using the parabolic coordinate $w = s(s-1)$ (which folds $s \leftrightarrow 1-s$) and identifying $h = w/c_{\mathrm{eff}}$ with a Kac table entry, the nontrivial zeros satisfy

$$t_n^2 = c_{\mathrm{eff}}\,|h_{r_n,s_n}(c_{\mathrm{eff}})| - \tfrac{1}{4}.$$

Conjecture IV.1 (Zeros from deformed Kac table)**.** *There exists a real $c_{\mathrm{eff}} > 2$ and a bijection $n \mapsto (r_n, s_n)$ such that the formula above recovers the exact Riemann zero heights.*

Problem IV.1 (Fixing c_{eff} algebraically)**.** Determine $c_{\mathrm{eff}} = 2\alpha$ in closed form from the Yang-Baxter recurrences. Candidate approaches: (A) Kac determinant fixed by first zero; (B) ratio $\sigma(2,0,0)/\sigma(0,0,0) = \pi^2/3$ as fixed-point equation in c; (C) modular weight $3/2$ of $\mathscr{H}(w)/P_0(w)$ uniquely determines c.

F. Spacetime topology and failure of real grading

The real grading proof (Step 2 of[1]) uses the boundary term $\tau(s,n,m) = [x^s \star B_n \star B_m]_0^\infty$ being real at both limits. In spacetime backgrounds that identify $\mathscr{I}^+$ with $\mathscr{I}^-$ (de Sitter, thermal AdS, Misner/Gödel, BMS), the boundary term acquires a complex phase and real grading fails.

Conjecture IV.2 (Topological failure)**.** *In spacetimes identifying conformal boundaries, the deformed ζ-analogue can have zeros off $\Re(s) = \frac{1}{2}$. The de Sitter case gives the quantitative prediction: deviation from realness of $\sigma_{dS}(2,0,0)$ is $(T_{dS}/m)^2$.*

V. VOLTERRA PRODUCT INTEGRALS, OPERATOR TRACES, AND ALGEBRAS OVER PRIMES

A. Vertical Volterra product integrals in the critical strip

Fix $x \in (0,1)$ and compactify the vertical ray $\{x + it : t \geq 0\}$ via $t = \tan\theta$, $\theta \in [0, \pi/2)$:

$$[0, \tfrac{\pi}{2}) \ni \theta \longmapsto \zeta(x + i\tan\theta) \in \mathbb{C} \cup \{\infty\}.$$

This maps each vertical half-line of the critical strip to a compact interval, sending $i\infty$ to the point at infinity of the Riemann sphere $\hat{\mathbb{C}}$. The image curve is a compactified trajectory of ζ that winds around 0 once for each zero on $\Re(s) = x$.

The *Volterra product integral* (product integral of the logarithmic derivative)

$$F(x) := \Re \int_0^{\pi/2} \frac{\zeta'(x + i\tan\theta)}{\zeta(x + i\tan\theta)} \frac{d\theta}{\cos^2\theta} = \Re \int_x^{x+i\infty} \frac{\zeta'(s)}{\zeta(s)}\, ds$$

counts zeros on the vertical line $\Re(s) = x$ via the argument principle.

Conjecture V.1 (Volterra detection of off-line zeros)**.** *$F(x)$ is real-analytic on $(0,1) \setminus \{1/2\}$, with logarithmic singularities precisely at values x_0 where ζ has a nontrivial zero on $\Re(s) = x_0$. In particular, RH $\Leftrightarrow$ F is smooth on $(0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$.*

Remark V.1 (Why this stalls)**.** The integral $\int_x^{x+i\infty} (\zeta'/\zeta)\, dt$ is only conditionally convergent. Proving $F(x_0) = 0$ for $x_0 \neq 1/2$ is equivalent to showing that the oscillations of ζ'/ζ along vertical lines cancel perfectly — which is essentially RH by another name. The geometric compactification $t = \tan\theta$ clarifies the topology but does not remove the analytic obstruction.

B. Infinite matrix traces and the von Neumann algebra

View $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ as the trace of the diagonal operator $D^{(s)} = \mathrm{diag}(n^{-s})_{n\geq 1}$ on $\ell^2(\mathbb{N})$:

$$\zeta(s) = \mathrm{Tr}(D^{(s)}).$$

The vertical product integral $F(x)$ is then the real part of the trace of $\partial_t \log M^{(x,t)}$ integrated over $t \geq 0$:

$$F(x) = -\Re \int_0^{\infty} \frac{d}{dt} \log \zeta(x+it)\, dt.$$

A zero of ζ at $x+it_0$ produces a pole of $\partial_t \log M^{(x,t)}$ at $t = t_0$, manifesting as a logarithmic divergence of the trace integral with residue equal to the zero's multiplicity.

C. Connection to Connes' trace formula

Connes[6] constructs a spectral triple $(\mathscr{A}, \mathscr{H}, \mathscr{D})$ with $\mathscr{H} = L^2(\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*)$ and a self-adjoint Dirac-type operator $\mathscr{D}$. The spectral action is

$$\mathrm{Tr}(f(\mathscr{D}/\Lambda)) \sim \Lambda \hat{f}(0)\zeta(1) - \sum_{\rho} \hat{f}(\rho) + O(1),$$

where the sum runs over nontrivial zeros ρ. The function $F(x)$ is a *partial trace* of the resolvent of $\mathscr{D}$, integrating out the imaginary direction. RH is the statement that this partial trace has no singular support off the single fiber $x = 1/2$.

The Dixmier trace $\mathrm{Tr}_{\omega}(T) = \lim_{\omega} \frac{1}{\log N} \sum_{n=1}^{N} \lambda_n(T)$, applied to $\mathscr{D}^{-1}$, recovers $\zeta(1)$ and its distributional prolongation. The vertical product integral at $x = 1$ is this Dixmier trace, regularized by $t = \tan\theta$.

D. Heisenberg algebra over primes: integers as words, primes as letters

Let $\mathscr{P} = \{2, 3, 5, 7, \ldots\}$ be the primes (an alphabet). Each integer $n = \prod_p p^{v_p(n)}$ is a word in the free commutative monoid $\mathrm{Mon}(\mathscr{P}) \cong (\mathbb{Z}_{>0}, \times)$.

Promote each prime p to bosonic creation/annihilation operators $a_p^{\dagger}, a_p$ on Fock space $\mathscr{F} = \bigoplus_{n\geq 0} \mathrm{Sym}^n(\ell^2(\mathscr{P}))$:

$$[a_p, a_q^{\dagger}] = \delta_{pq}, \qquad [a_p, a_q] = [a_p^{\dagger}, a_q^{\dagger}] = 0.$$

Each integer n becomes the Fock state $|n\rangle = \prod_p (a_p^{\dagger})^{v_p(n)} / \sqrt{v_p(n)!}\, |0\rangle$. The unique factorization theorem is the Poincaré-Birkhoff-Witt theorem for the commutative enveloping algebra: every basis state $|n\rangle$ is unique.

The *logarithmic number operator*

$$H = \sum_p \log(p)\, a_p^{\dagger} a_p$$

satisfies $H|n\rangle = \log(n)|n\rangle$, and its partition function is exactly ζ:

$$\mathrm{Tr}(e^{-\beta H}) = \sum_{n=1}^{\infty} n^{-\beta} = \zeta(\beta), \qquad \beta > 1.$$

Remark V.2 (Primes as left-moving modes)**.** In a 2d CFT, the Virasoro character $\mathrm{Tr}_{\mathscr{H}}(q^{L_0})$ counts operator states built from mode applications $L_{-n_1} \cdots L_{-n_k}|h\rangle$. Here $\mathrm{Tr}(q^H) = \zeta(-\log q)$ counts integer states built from prime modes $a_p^{\dagger}$. Primes are the simple roots of the commutative mode algebra; unique factorization is PBW.

E. The center of the Heisenberg algebra and Hecke operators

Operators commuting with all mode products $a_p^{\dagger} a_q$ are exactly the *multiplicative arithmetic functions* $f : \mathbb{Z}_{>0} \to \mathbb{C}$ (diagonal operators $T = \sum_n f(n)|n\rangle\langle n|$ with $f(mn) = f(m)f(n)$ for $\gcd(m,n) = 1$). Their Dirichlet series $\sum f(n) n^{-s}$ are the Euler-product L-functions.

The Hecke operators $\{T_p\}$ generate a commutative (Hecke) algebra whose center consists of spherical functions — automorphic L-functions for $\mathrm{GL}(1)$ (Dirichlet characters) and $\mathrm{GL}(2)$ (classical modular forms).

F. Connection to the Langlands program

The four-floor algebraic tower is:

1. **Arithmetic** ($\mathbb{Z}$, $(\mathbb{Z}_{>0}, \times)$): commutative, primes generate.

2. **Heisenberg** (CCR over $\mathscr{P}$): non-commutative in creation/annihilation; diagonal (abelian) in the number basis. Center = multiplicative arithmetic functions.

3. **Hecke algebra**: acts on Hilbert space of arithmetic functions; automorphic side of Langlands for $\mathrm{GL}(1)$.

4. **Virasoro / W_{∞}**: modes of $T(z) =: (\partial\phi)^2 :$ built from the free boson on the charge lattice $(\log \mathscr{P}) \subset \mathbb{R}$. Central charge $c = c_{\mathrm{eff}} = 2\alpha$ connects back to §IV.

Each floor is a non-abelian extension of the one below. RH on the ground floor translates to: the spectrum of L_0 (Virasoro zero mode, fourth floor) has all eigenvalues $h = h_{r,s}$ real and positive — the Kac-table condition of §IV.

The Bost-Connes algebra[7] $\mathscr{A}_{\mathrm{BC}}$ (with generators μ_n, $e(r)$, $r \in \mathbb{Q}/\mathbb{Z}$) provides the C*-algebra bridge between floors 2 and 3: its KMS states at inverse temperature $\beta > 1$ give $\zeta(\beta)$ as partition function, the same ζ appearing in the Fock-space trace at level 2.

Conjecture V.2 (Center conjecture)**.** *The center of the Virasoro algebra acting on $\mathscr{F}$ is generated by the Casimir $C = L_0 - c/24$. Reality of the spectrum of C is equivalent to $c_{\mathrm{eff}} \in \mathbb{R}$, which is the real-grading condition of §IV, which implies RH.*

VI. STRUCTURAL SIMILARITIES: ONE THEOREM IN FOUR LANGUAGES

All four approaches are the same meta-theorem in different languages. We make the dictionary explicit.

A. The common spectral operator and its involution

Each approach defines an operator $\mathscr{O}$ whose spectrum (or whose kernel, or whose resolvent singularities) encodes the zeros of ζ:

Weyl derivative. $\mathscr{O} = \partial^s$, a multiplicative operator on Fourier modes. The critical line is the locus where the even/odd symmetry $a_n = \pm a_{-n}$ holds, i.e., where the involution $f(\theta) \mapsto f(-\theta)$ is compatible with $\partial^s f = 0$.

Schrödinger–Riemann. $\mathscr{O} = -\frac{1}{2}\partial_x^2$ acting on $[0,\pi]$. The involution is $x \mapsto \pi - x$ (spatial reflection), which enforces $A_n = A_{-n}$ and confines the "forbidden normalization energies" to $\Re(s) = 1/2$.

Yang-Baxter / Virasoro. $\mathscr{O} = L_0$ (Virasoro zero mode). The involution is $s \mapsto 1-s$ (functional equation symmetry), encoded in the Yang-Baxter system by the parabolic coordinate $w = s(s-1)$ which is *invariant* under $s \mapsto 1-s$. The real grading $\alpha \in \mathbb{R}$ is the fixed-point condition under this involution.

Volterra / Langlands. $\mathscr{O} = \mathscr{D}$ (Connes Dirac) or $H = \sum_p \log p\, a_p^\dagger a_p$. The involution is adelic duality $\mathbb{A}_\mathbb{Q}/\mathbb{Q}^* \to (\mathbb{A}_\mathbb{Q}/\mathbb{Q}^*)^\vee$, which corresponds to $n \mapsto 1/n$ on the diagonal matrices, i.e., $s \mapsto 1-s$ on the Mellin transform.

B. The symmetry condition in each language

Approach	Symmetry / reality condition
Weyl derivative	$a_n = a_{-n}$: real (cosine) Fourier series
Schrödinger	$A_n = A_{-n}$: spatially symmetric wave function
Yang-Baxter	$\alpha \in \mathbb{R}$: real decay exponent / real grading of IBP system
Connes / Langlands	$F(x)$ real-analytic off $x = \frac{1}{2}$; real spectrum of L_0

In every case the condition is: some distinguished real-to-complex map is real along the orbit of the natural involution $s \mapsto 1-s$.

C. The Weyl derivative modes are Virasoro generators

The Witt mode $d_n = -e^{in\theta}\partial_\theta$ is the n-th Fourier mode of ∂_θ, and $\partial^s = \sum_n (in)^s a_n e^{in\theta}$ is the s-th complex power of ∂_θ. On the Fock space $\mathscr{F}$ of §V, ∂_θ corresponds to the generator L_{-1} (translation), and $(in)^s$ corresponds to the spectral weight n^s of the mode. The Weyl decomposition (1) is then:

$$\partial^s f = \underbrace{i^s L(-s,\chi(n))}_{\text{left-moving / holomorphic}} + \underbrace{(-i)^s L(-s,\chi(-n))}_{\text{right-moving / anti-holomorphic}} ,$$

the chiral decomposition of the fractional derivative into holomorphic ($a_n e^{in\theta}$) and anti-holomorphic ($a_{-n}e^{-in\theta}$) sectors. Disjointness of Z_+ and Z_- (Conjecture II.1) is the condition that the holomorphic and anti-holomorphic sectors have no common zero, which is a version of the statement that the CFT has no operator at dimension $h = \bar{h}$ except on the diagonal — reflecting the real structure of the operator algebra.

D. Normalization and the partition function

The Schrödinger normalization condition $\|\psi\|_{L^2} = 1$ and the Yang-Baxter partition function $\mathrm{Tr}(e^{-\beta H}) = \zeta(\beta)$ are the same object: both are the L^2 norm squared of a wave function (or a density-matrix trace) in the respective Hilbert space. The "forbidden normalization energies" of Conjecture III.1 are the zeros of ζ; the "spectral zeta function" of the Schrödinger Hamiltonian is $\zeta(s) = \sum_n E_n^{-s}$ by construction.

VII. GENERALIZATIONS OF ζ VIA SEED AND QUASI-PERIODICITY

The Yang-Baxter system has two free inputs: the seed integral and the quasi-periodicity factor. Varying them generates a natural family generalizing ζ.

A. Varying the seed kernel

Replace the Fermi-Dirac kernel $K(x) = e^x(e^x+1)^{-2}$ by any K that is smooth on $(0,\infty)$, rapidly decaying at $+\infty$, and controlled at 0. Representative examples:

- $K(x) = x^a e^{-x}$: seeds $\Gamma(s+a+1)$.
- $K(x) = e^x(e^x-1)^{-2}$ (Bose kernel): same $q = 1/4$ quasi-periodicity but different functional equation.
- $K(x) = e^x(e^x+1)^{-n}$, $n \geq 2$: higher-order Fermi kernels, $q = 4^{-(n-1)}$.
- $K(x) = J_\nu(x)e^{-x}$: Bessel kernels, Mellin transform related to L-functions of elliptic curves.

The seed determines the Euler product: $K(px)/K(x) = p^{-s}$ for all primes p is the multiplicativity condition; its unique solution (up to normalization) is the Fermi-Dirac kernel, explaining why ζ occupies a distinguished position.

In the Weyl framework, changing the seed corresponds to changing the Fourier coefficients $a_n = \hat{K}(n)$: different kernels give different Dirichlet series $L(-s,\chi(n))$, and the zero-set disjointness question becomes theory-dependent.

B. Varying the quasi-periodicity factor

Replace $q = 1/4$ by $q \in (0,1)$. The quasi-periodicity becomes $\sigma^{(q)}(s,n+2,m) = q\,\sigma^{(q)}(s,n,m)$. The Coulomb-gas

identification becomes $\alpha_+^{(q)}\alpha_-^{(q)} = q$, with the deformation parameter $q_{\mathrm{YB}} = \sqrt{q}$ of the quantum group $U_q(\mathfrak{sl}(2))$.

Conjecture VII.1 (Family of L-functions)**.** *For each $q \in (0,1)$ there exists an L-function $L^{(q)}(s)$ satisfying RH if and only if the Yang-Baxter quadratic coefficients $a_{2j,2k}^{(q)}$ are real. The family interpolates: $L^{(1/4)} = \zeta$, $L^{(q)} \to 1$ as $q \to 0$.*

VIII. OPEN PROBLEMS

Problem VIII.1 (Algebraic determination of c_{eff})**.** Determine $c_{\mathrm{eff}} = 2\alpha \approx 2.220$ in closed form from the Yang-Baxter recurrences. See Problem IV.1 for candidate approaches.

Problem VIII.2 (Kac label map)**.** Find an explicit bijection $n \mapsto (r_n, s_n) \in \mathbb{N}^2$ such that $t_n^2 = c_{\mathrm{eff}}|h_{r_n,s_n}| - 1/4$ holds for all nontrivial zeros.

Problem VIII.3 (Proving disjointness of zero sets)**.** Prove (or disprove) Conjecture II.1: for generic $\{a_n\}$, $Z_+ \cap Z_- = \emptyset$. Even a proof in the case $a_n = n^{-\sigma}$ (standard Dirichlet series) would be interesting.

Problem VIII.4 (Volterra conditional convergence)**.** Find a regularization of the vertical product integral $F(x)$ that (a) is absolutely convergent and (b) still detects off-line zeros, without presupposing RH.

Problem VIII.5 (Langlands center condition)**.** Make Conjecture V.2 precise: identify the center of the Virasoro algebra acting on $\mathscr{F}$ and show that real spectrum of L_0 implies RH, without a circular argument.

Problem VIII.6 (De Sitter zeta analogue)**.** Compute $\sigma_{dS}(2,0,0)$ perturbatively in T_{dS}/m and verify Conjecture IV.2 quantitatively.

IX. SUMMARY

Approach	Main tool	Symmetry condition	Status
Weyl derivative	$\partial^s = i^s L(-s,\chi(n)) + (-i)^s L(-s,\chi(-n))$	$Z_+ \cap Z_- = \emptyset \Rightarrow a_n = a_{-n}$	Conjecture
Schrödinger	Forbidden normalization; $\lvert D(s)\rvert^2 = 0$	L^2 symmetry; $A_n = A_{-n}$	Heuristic
Yang-Baxter	Real grading; quadratic $as^2 - as + 1$	$\alpha \in \mathbb{R}$, $a < 4$	**Complete**[1]
Volterra / Langlands	Arg. principle; Connes trace; Fock $\mathscr{F}$	Partial trace smooth off $x = \frac{1}{2}$	Analytic gap

The Yang-Baxter quadratic-factor argument is the only approach that has been closed into a rigorous proof. The others remain as conjectures or heuristics, each illuminating a different face of the same spectral reality condition that forces zeros onto the critical line.

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